High-dimensional Asymptotics of Feature Learning: How One Gradient Step Improves the Representation

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Introduction: Two-Layer Neural Network (NN)

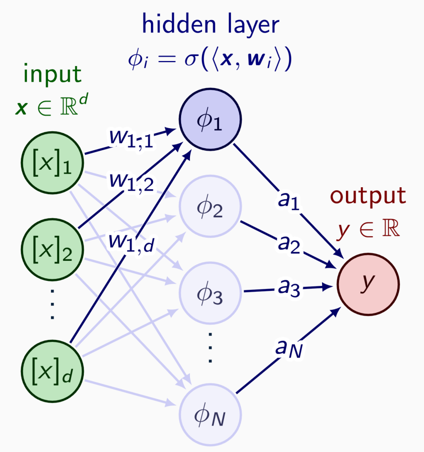
Width-N Two-Layer NN

$$f_{\mathrm{NN}}(\mathbf{x}) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} a_i \, \sigma(\mathbf{x}^\top \mathbf{w}_i) = \frac{1}{\sqrt{N}} \mathbf{a}^\top \sigma(\mathbf{W}^\top \mathbf{x}).$$

- Input data: $\mathbf{x} \in \mathbb{R}^d$.
- Trainable parameters: $\mathbf{W} \in \mathbb{R}^{d \times N}$, $\mathbf{a} \in \mathbb{R}^N$.
- Element-wise nonlinearity: $\sigma : \mathbb{R} \to \mathbb{R}$.

Optimization: given a convex loss ℓ,

- Optimizing $\mathbf{a}$ under fixed $\mathbf{W}$ is *convex*.
- Optimizing $\mathbf{W}$ under fixed $\mathbf{a}$ is *non-convex*.



Our Goal: precise characterization of the performance of the trained NN.

Introduction: Training and Test Setting

- **Training.** Empirical risk minimization (potentially ℓ_2-regularized):

$$\mathcal{L}(f) = \frac{1}{n}\sum_{i=1}^{n}(f(x_i) - y_i)^2, \quad y_i = f^*(x_i) + \varepsilon_i,$$

  where f^* is the target function (teacher model), and ε is i.i.d. label noise.

- **Test.** Prediction risk: $R(f) = \mathbb{E}_x[(f(x) - f^*(x))^2] = \|f - f^*\|^2_{L^2(P_x)}$.

Regime of Interest – Proportional Asymptotic Limit

$n, d, N \to \infty, \quad n/d \to \psi_1, \quad N/d \to \psi_2,$ where $\psi_1, \psi_2 \in (0, \infty)$.

Why is this an interesting regime to analyze?

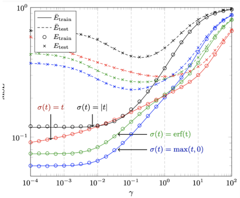
- It corresponds to the setting where the network width and data size are comparable, consistent with practical choices of model scaling.
- It might be possible to derive the *precise* prediction risk in this limit.

Kernel Models Related to NN

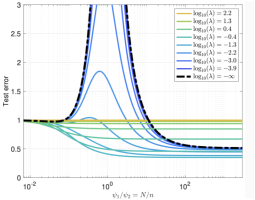
Two widely-studied kernels derived from a two-layer NN:

- **Conjugate Kernel (CK)** with features: $\phi_{\mathrm{CK}}(\mathbf{x}) = \frac{1}{\sqrt{N}}\sigma(\mathbf{W}^\top \mathbf{x}) \in \mathbb{R}^N$.
  Regression on the CK corresponds to fixing $\mathbf{W}$ and only learning the 2nd layer $\mathbf{a}$.
- **Tangent Kernel (NTK)** with features: $\phi_{\mathrm{NT}}(\mathbf{x}) = \frac{1}{\sqrt{Nd}}\,\mathrm{Vec}\big(\sigma'(\mathbf{W}^\top \mathbf{x})\,\mathbf{x}^\top\big) \in \mathbb{R}^{Nd}$.
  This kernel arises from gradient descent on certain wide neural networks.

When $\mathbf{W}$ is randomly initialized, we arrive at a **random features (RF)** model, the precise asymptotics of which has been extensively studied in the proportional limit.



[Louart, Liao, and Couillet, 2018].



[Mei and Montanari, 2019].

Limitation of Kernel Ridge Regression

Can these RF models fully capture the effectiveness of NNs? *Not quite...*

Consider the *ridge regression estimator* for $\mathrm{RF} \in \{\mathrm{CK}, \mathrm{NT}\}$:

$$f_{\mathrm{RF}}^{\lambda}(\mathbf{x}) = \langle \phi_{\mathrm{RF}}(\mathbf{x}), \hat{\mathbf{a}}^{\lambda} \rangle, \qquad \hat{\mathbf{a}}_{\lambda} = \arg\min_{\mathbf{a}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \big(y_i - \langle \phi_{\mathrm{RF}}(\mathbf{x}_i), \mathbf{a} \rangle \big)^2 + \frac{\lambda}{N} \|\mathbf{a}\|_2^2 \right\}.$$

Theorem (Ghorbani et al. 19, Hu and Lu 20, Bartlett et al. 21, . . .)

[Informal] Denote $P_{>1}$ as the projector orthogonal to constants and linear functions in $L^2(P_X)$. Then under certain concentration conditions on the input $\mathbf{x}$, we have[1]

$$\inf_{\lambda > 0} \min\{\mathcal{R}(f_{\mathrm{CK}}^{\lambda}), \mathcal{R}(f_{\mathrm{NT}}^{\lambda})\} \ \geq\ \boxed{\|P_{>1} f^*\|_{L^2}^2} + o_{d,\mathbb{P}}(1),$$

- In the proportional limit, RF models can only learn **linear functions**.
- NNs are clearly more powerful than linear models on the input...

[1] Similar lower bound also holds for certain rotationally invariant kernels studied in [El Karoui 10].

Feature Learning in Two-Layer NN

Where does this gap come from? **Feature Learning!**

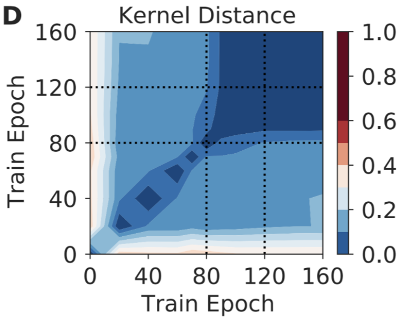
- When we optimize the first-layer parameters $\mathbf{W}$, we expect the model to "adapt" to the data and learn useful representations.
- In RF models, $\mathbf{W}$ is fixed, so there is no "representation learning".

Motivation: Can we precisely capture the presence of *feature learning* in the proportional limit, when the first-layer $\mathbf{W}$ is optimized via *gradient descent*?

Empirical Observation:

- Neural network features often change most rapidly in the **early phase** of gradient descent (GD) training.

We consider the most simplified setting of the "early phase": one gradient step on $\mathbf{W}$, and analyze how the learned CK adapts to the learning problem.

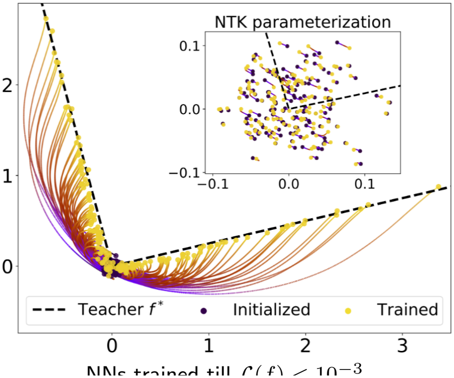


Problem Setting: Basic Assumptions

1. **Proportional Limit.** $n, d, N \to \infty, \quad \mathbf{n/d} \to \psi_1, \quad \mathbf{N/d} \to \psi_2, \quad \psi_1, \psi_2 \in (0, \infty)$.
2. **Student-teacher Setup.** $y_i = f^*(\mathbf{x}_i) + \varepsilon_i$, where $\mathbf{x}_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbf{I})$, ε_i is i.i.d. noise with variance σ_ε^2, and f^* is Lipschitz with $\|f^*\|_{L^2} = \Theta_d(1)$.
3. **Normalized Activation.** σ has bounded first three derivatives, and is normalized such that $\mathbb{E}[\sigma(z)] = 0$, $\mathbb{E}[z\sigma(z)] = \mu_1 \neq 0$, for $z \sim \mathcal{N}(0, 1)$.
4. **Gaussian Initialization.** $[\mathbf{W}_0]_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/d), \quad [\mathbf{a}]_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/N)$.

Note: we use the mean-field parameterization, which admits a *feature learning limit* (i.e., the weights do not "freeze" around the initialization).

$$f(\mathbf{x}) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} a_i \sigma(\langle \mathbf{x}, \mathbf{w}_i \rangle) = \frac{1}{\sqrt{N}} \underbrace{\mathbf{a}^\top}_{\approx 1/N} \sigma(\mathbf{W}^\top \mathbf{x}).$$



Problem Setting: One-step Gradient Descent

- **One-step GD on 1st Layer.** We take one **gradient step**[3] on the empirical MSE loss $\mathcal{L}(f) = \frac{1}{n}\sum_{i=1}^{n}(f(\mathbf{x}_i) - y_i)^2$, that is, $\mathbf{W}_1 = \mathbf{W}_0 + \eta\sqrt{N}\cdot\mathbf{G}_0$, where

$$\mathbf{G}_0 := -\frac{1}{n}\mathbf{X}^\top\left[\left(\frac{1}{\sqrt{N}}\sigma(\mathbf{X}\mathbf{W}_0)\mathbf{a} - \mathbf{y}\right)\mathbf{a}^\top \odot \sigma'(\mathbf{X}\mathbf{W}_0)\right],$$

- **Ridge Regression for 2nd Layer.** After learning features for one step, we perform ridge regression on the trained CK using **a fresh set of data** $\{\tilde{\mathbf{X}}, \tilde{\mathbf{y}}\}$:

$$\hat{\mathbf{a}}_\lambda = \arg\min_{\mathbf{a}}\left\{\frac{1}{n}\|\tilde{\mathbf{y}} - \Phi\mathbf{a}\|^2 + \frac{\lambda}{N}\|\mathbf{a}\|^2\right\}, \qquad \Phi := \frac{1}{\sqrt{N}}\sigma(\tilde{\mathbf{X}}\mathbf{W}_1) \in \mathbb{R}^{n\times N}.$$

Denote $f_{\mathrm{GD}}^{\lambda}(\mathbf{x}) = \frac{1}{\sqrt{N}}\hat{\mathbf{a}}_\lambda^\top\sigma(\mathbf{W}_1^\top\mathbf{x})$, and prediction risk: $\mathcal{R}_{\mathrm{GD}}(\lambda) = \mathcal{R}(f_{\mathrm{GD}}^{\lambda})$.

This Work: Aim to compute $\mathcal{R}_{\mathrm{GD}}(\lambda)$, and show its *improvement* over the initialized RF, and potentially over the *lower bound* $\|P_{>1}f^*\|_{L^2}^2$.

Challenge: cannot directly use *random matrix theory*, as $\mathbf{W}_1$ is no longer "random".

Properties of the Gradient Matrix $\mathbf{G}_0$

Can we exploit certain structure of the first GD step to simplify the calculation?

Orthogonal Decomposition of σ:

$\sigma(z) = \mu_1 z + \sigma_\perp(z)$, where $\mu_1 = \mathbb{E}[\sigma'(z)] \;\Rightarrow\; \mathbb{E}[\sigma_\perp(z)] = \mathbb{E}[z\sigma_\perp(z)] = 0.$

Proposition (BES+22)

Recall $\mathbf{G}_0 = \frac{1}{\eta\sqrt{N}}(\mathbf{W}_1 - \mathbf{W}_0)$. Define rank-1 matrix $\mathbf{A} := \frac{\mu_1}{n\sqrt{N}}\mathbf{X}^\top \mathbf{y}\,\mathbf{a}^\top$. Then

$$\sqrt{N} \cdot \|\mathbf{G}_0 - \mathbf{A}\| \lesssim \|\mathbf{G}_0\|, \quad \text{w.h.p.}$$

Intuition: Many commonly-used activations are monotone, so σ' is not centered:

$$n\sqrt{N} \cdot \mathbf{G}_0 = \mu_1 \mathbf{X}^\top(\mathbf{y} - f_0(\mathbf{X}))\mathbf{a}^\top + \mathbf{X}^\top\left[(\mathbf{y} - f_0(\mathbf{X}))\mathbf{a}^\top \odot \sigma'_\perp(\mathbf{X}\mathbf{W}_0)\right]$$

Hence $\mathbf{G}_0$ contains: $\|\mathbf{A}\|_F \asymp \|\mathbf{B}\|_F$, but $\|\mathbf{A}\| \gg \|\mathbf{B}\|$:

- A rank-1 "spike" $\mathbf{A}$
- A "residual" with smaller operator norm (but not Frobenius norm) $\mathbf{B}$

Selection of Learning Rate η

Based on the decomposition of $\mathbf{G}_0$, we focus on the following choices[4] of η:

- Small lr: $\eta = \Theta(1) \;\Rightarrow\; \|\mathbf{W}_1 - \mathbf{W}_0\| \asymp \|\mathbf{W}_0\|$.
- Large lr: $\eta = \Theta(\sqrt{N}) \;\Rightarrow\; \|\mathbf{W}_1 - \mathbf{W}_0\|_F \asymp \|\mathbf{W}_0\|_F$.

Remarks:

- Under $\eta = \Theta(1)$, the NN after one GD step remains close to the **kernel regime**: each neuron (or parameter) does not travel far away from the initialization, i.e., $|[\mathbf{W}_1 - \mathbf{W}_0]_{ij}| \ll |[\mathbf{W}_0]_{ij}|$ for all i, j with high probability.
- $\eta = \Theta(\sqrt{N})$ mirrors the **maximal update parameterization** [Yang and Hu 2020]: for $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I})$, the change in each coordinate of the feature vector is significant, i.e., $|\sigma(\mathbf{W}_1^\top \mathbf{x})_i - \sigma(\mathbf{W}_0^\top \mathbf{x})_i| \asymp |\sigma(\mathbf{W}_0^\top \mathbf{x})_i| = \tilde{\Theta}(1)$ for all i with high probability.

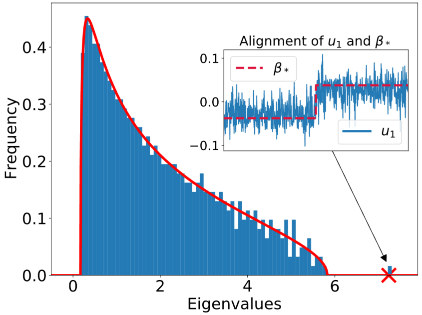
A Spiked Model for W^1

- $\sigma = \tanh$, $f^*(x) = \mathrm{ReLU}(\langle x, \beta^* \rangle)$.
- Teacher vector $\beta^* \propto [-\mathbf{1}_{d/2}; \mathbf{1}_{d/2}]$.
- $\psi_1 = n/d = 4, \quad \psi_2 = N/d = 2$.
- $\eta = 2$.

Observation: after one gradient step with learning rate $\eta = \Theta(1)$:

- The **bulk** of the spectrum of W remains unchanged. (Initialized W^0 spectrum $\sim$ Marchenko–Pastur law.)
- A **spike** ($\times$) appears in W^1, which *aligns with the linear component of f^**.

Blue: empirical simulation. Red: analytic prediction (next slide).



Spiked Model for W_1 (continued)

Orthogonal Decomposition: $f^*(\mathbf{x}) = \mu_0^* + \mu_1^*\langle \mathbf{x}, \boldsymbol{\beta}_* \rangle + \mathrm{P}_{>1} f^*(\mathbf{x})$,

- *Linear part*: $\|\boldsymbol{\beta}_*\| = 1$, $\mu_1^* \boldsymbol{\beta}_* = \mathbb{E}[\mathbf{x} f^*(\mathbf{x})]$; *Nonlinear part*: $\|\mathrm{P}_{>1} f^*\|_{L^2} = \mu_2^*$.

Theorem (BES+22)

For $\eta = \Theta(1)$, *define* $\theta_1 := \sqrt{\|f^*\|_{L^2}^2 \psi_1^{-1} + {\mu_1^*}^2} \cdot \mu_1 \eta$, $\theta_2 := \mu_1 \mu_1^* \eta$. *The leading singular value* $s_1(\mathbf{W}_1)$ *and the corresponding singular vector* $\mathbf{u}_1$ *satisfy*

$$s_1(\mathbf{W}_1) \to \sqrt{\frac{(1+\theta_1^2)(\psi_2+\theta_1^2)}{\theta_1^2}}, \qquad |\langle \mathbf{u}_1, \boldsymbol{\beta}_* \rangle|^2 \to \frac{\theta_2^2}{\theta_1^2}\left(1 - \frac{\psi_2 + \theta_1^2}{\theta_1^2(\theta_1^2+1)}\right),$$

for $\theta_1 > \psi_2^{1/4}$; *otherwise*, $s_1(\mathbf{W}_1) \to 1 + \sqrt{\psi_2}$, $|\langle \mathbf{u}_1, \boldsymbol{\beta}_* \rangle| \to 0$.

When η exceeds some threshold, a "spike" appears:

- Increase step size η $\Rightarrow$ <u>larger spike $s_1(\mathbf{W}_1)$</u>.
- Increase sample size ψ_1 $\Rightarrow$ <u>greater alignment</u>.

A Spiked Model for CK?

Question: How does the spike in $\mathbf{W}_1$ affect the *kernel (CK) matrix*?

For $\eta = \Theta(1)$, and *odd activation* σ, the expected CK matrix $\bar{\Sigma}_\Phi$ satisfies

$$\hat{\Sigma}_\Phi - \bar{\Sigma}_\Phi \xrightarrow{\mathbb{P}} 0, \qquad \bar{\Sigma}_\Phi = \mathbb{E}_{\mathbf{x}}\left[\sigma(\mathbf{W}_1^\top \mathbf{x})\sigma(\mathbf{x}^\top \mathbf{W}_1)\right] = \mu_1^2 \mathbf{W}_1^\top \mathbf{W}_1 + \mu_2^2 \mathbf{I}.$$

- Intuitively, we expect a spike to appear in the (empirical) CK matrix.
- How do we predict properties of the CK spike? **Gaussian Equivalence**

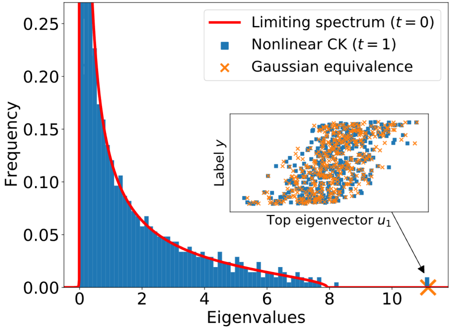
Nonlinear CK : $\Phi = \frac{1}{\sqrt{N}}\sigma(\tilde{\mathbf{X}}\mathbf{W}_1)$, "Linearized" CK: $\bar{\Phi} = \frac{1}{\sqrt{N}}\left(\mu_1 \tilde{\mathbf{X}}\mathbf{W}_1 + \mu_2 \mathbf{Z}\right)$.

Conjecture (Gaussian Equivalence of CK Spike)

For odd activation σ and $\eta = \Theta(1)$, given i.i.d. training data $\tilde{\mathbf{X}}, \tilde{\mathbf{y}}$ (independent of $\mathbf{W}_1$). Denote the left singular vectors of $\Phi, \bar{\Phi}$ as $\mathbf{u}_1, \bar{\mathbf{u}}_1$, we conjecture

$$\left|s_i(\Phi) - s_i(\bar{\Phi})\right| = o_{d,\mathbb{P}}(1),\ \forall i \in [n]; \qquad \left|\langle \mathbf{u}_1, \tilde{\mathbf{y}}/\left\|\tilde{\mathbf{y}}\right\|\rangle\right|^2 = \left|\langle \bar{\mathbf{u}}_1, \tilde{\mathbf{y}}/\left\|\tilde{\mathbf{y}}\right\|\rangle\right|^2 + o_{d,\mathbb{P}}(1).$$

Spiked Model for CK (continued)



Blue: empirical simulation.
Red: analytic prediction (initial CK).
Orange: Gaussian equivalence.

- $\sigma = \mathrm{SoftPlus}$.
- $f^*(\mathbf{x}) = \tanh(\langle \mathbf{x}, \boldsymbol{\beta}_* \rangle)$.
- $\psi_1 = n/d = 3/2, \ \psi_2 = N/d = 5/4$.
- $\eta = 2$.

- The **bulk** of the CK spectrum remains unchanged [6].
- A **spike** ($\times$) appears in the learned CK, predicted by Gaussian equivalence.
- The corresponding eigenvector $\mathbf{u}_1$ aligns with training labels $\tilde{\mathbf{y}}$.

[6] The spectrum of the initialized CK_0 is characterized in [Fan and Wang 2020].

Prediction Risk of CK Ridge Regression

Question: does this alignment improve the performance of the kernel model?

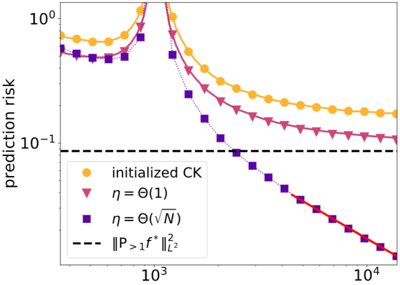
Case Study: Single-index target[7]. $f^*(\mathbf{x}) = \sigma_*(\langle \mathbf{x}, \boldsymbol{\beta}_* \rangle)$, where $\|\boldsymbol{\beta}_*\| = 1$, and σ_* is Lipschitz with $\mu_0^* = 0$, $\mu_1^* \neq 0$.

Goal: compute the prediction risk $\mathcal{R}_{\mathrm{GD}}(\lambda)$ of the ridge estimator

$$f_{\mathrm{GD}}^{\lambda}(\mathbf{x}) = \frac{1}{\sqrt{N}} \hat{\mathbf{a}}_\lambda^\top \sigma\left(\mathbf{W}_1^\top \mathbf{x}\right), \quad \hat{\mathbf{a}}_\lambda = \arg\min_{\mathbf{a}} \left\{ \frac{1}{n} \left\| \tilde{\mathbf{y}} - \tfrac{1}{\sqrt{N}} \sigma(\tilde{\mathbf{X}} \mathbf{W}_1) \mathbf{a} \right\|^2 + \frac{\lambda}{N} \|\mathbf{a}\|^2 \right\}.$$

We consider the following learning rate scalings:

- Small lr $\eta = \Theta(1)$: trained CK always improve upon the initial CK ridge estimator ($\mathcal{R}_0(\lambda)$).
- Large lr $\eta = \Theta(\sqrt{d})$: for some f^*, trained CK may outperform the lower bound $\|\mathrm{P}_{>1} f^*\|_{L^2}$.



[7] This setting is often studied in RF regression (e.g. [Gerace et al. 20],[Dhifallah and Lu 20]).

The Gaussian Equivalence Property

Consider the prediction risk of ridge regression on features $F \in \{\mathrm{CK}, \mathrm{GE}\}$:

$$\mathcal{R}_F(\lambda) = \mathbb{E}_{\mathbf{x}}\Big(\langle \phi_F(\mathbf{x}), \hat{\mathbf{a}}_\lambda \rangle - f^*(\mathbf{x})\Big)^2, \quad \hat{\mathbf{a}}_\lambda = \arg\min_{\mathbf{a}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (y_i - \langle \phi_F(\mathbf{x}_i), \mathbf{a} \rangle)^2 + \frac{\lambda}{N} \|\mathbf{a}\|^2 \right\}$$

- CK (nonlinear) : $\phi_{\mathrm{CK}}(\mathbf{x}) = \frac{1}{\sqrt{N}} \sigma(\mathbf{W}^\top \mathbf{x})$.
- GE (linear) : $\phi_{\mathrm{GE}}(\mathbf{x}) = \frac{1}{\sqrt{N}} (\mu_1 \mathbf{W}^\top \mathbf{x} + \mu_2 \mathbf{z})$, $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$,
  where $\mu_1 = \mathbb{E}[z\sigma(z)]$, $\mu_2 = \sqrt{\mathbb{E}[\sigma(z)^2] - \mu_1^2}$.

The **Gaussian Equivalence Property** refers to: $\mathcal{R}_{\mathrm{CK}}(\lambda) \approx \mathcal{R}_{\mathrm{GE}}(\lambda)$.

Previously, the GET has been shown for certain RF models [Hu and Lu 2020], but not for the trained features.

Implications of the Gaussian Equivalence:

- We can equivalently compute $\mathcal{R}_{\mathrm{GE}}$, which can be handled via RMT tools ☺
- The nonlinear CK model achieves the same performance as a linear model ☹

Gaussian Equivalence for Trained Features

Theorem (BES+22)

Assume σ is **odd** *in addition to the previous assumptions, then for fixed $t \in \mathbb{N}$, after the first-layer $\mathbf{W}$ is trained for t gradient steps with $\eta = \Theta(1)$,*

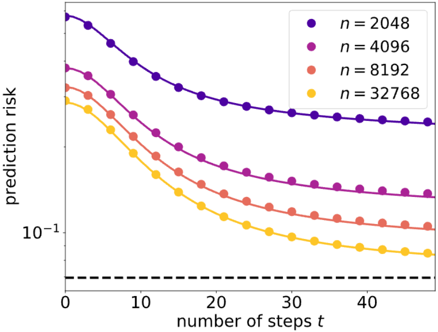
$$|\mathcal{R}_{\mathrm{CK}}(\lambda) - \mathcal{R}_{\mathrm{GE}}(\lambda)| = o_{d,\mathbb{P}}(1), \quad \textit{for } \lambda > 0.$$

Intuition: GET holds when $\mathbf{W}_t$ is not far away from the random initialization $\mathbf{W}_0$.

Figure: dots represent empirical values, solid curves are asymptotics predicted by CGMT.

- For learning rate $\eta = \Theta(1)$, GET remains accurate in the early phase of training.
- Prediction risk $\mathcal{R}_{\mathrm{GD}}(\lambda)$ can improve, but is still lower-bounded by $\|\mathrm{P}_{>1} f^*\|_{L^2}^2$

$\sigma = \mathrm{ReLU}, \ \sigma^* = \tanh.$



Gaussian Equivalence Theorem (Continued)

Proof Sketch. We extend the argument in [Hu and Lu 2020]:

1. Lindeberg exchange. Let $\hat{g}^k$ be the solution of the optimization problem:

$$L_k \triangleq \min_{g \in \mathbb{R}^N} \left(\sum_{i=1}^{k} \ell(y_i, \langle g, \phi_{\mathrm{GE}}(x_i) \rangle) + \sum_{j=k+1}^{n} \ell(y_j, \langle g, \phi_{\mathrm{CK}}(x_j) \rangle) + \frac{n\lambda}{N} \|g\|_2^2 + Q(g) \right)$$

As there are N total swaps, it suffices to show that for bounded test function ζ,

$$\mathbb{E}\zeta\left(\tfrac{1}{N} L_k\right) - \mathbb{E}\zeta\left(\tfrac{1}{N} L_{k-1}\right) = O\left(\frac{\operatorname{polylog} N}{N^{3/2}}\right). \tag{A}$$

2. Central limit theorem. A crucial step in establishing (A) is the CLT:

$$\mathbb{E}\varphi(\langle \phi_{\mathrm{GE}}, g \rangle) - \mathbb{E}\varphi(\langle \phi_{\mathrm{CK}}, g \rangle) = O\left(\frac{\operatorname{polylog} N}{\sqrt{N}} \cdot (1 + \|g\|_\infty^2)\right).$$

This is shown using Stein's method, when W has near-orthogonal columns.

3. ℓ^∞-norm control. Finally, we show that entries of $\hat{g}^k$ are "evenly distributed":

$$\Pr\left(\left\|\hat{g}^k\right\|_\infty \geq \operatorname{polylog} N\right) \leq \exp(-c \log^2 N), \quad \forall k \in [N].$$

Analysis of Small Learning Rate ($\eta = \Theta(1)$)

Goal: can we rigorously show that one feature learning step always *decreases* the prediction risk of the CK ridge regression estimator?

- Risk of *initial* CK (random features): $\mathcal{R}_0(\lambda) = \mathbb{E}_{\mathbf{x}}\big(\langle \sigma(\mathbf{W}_0^\top \mathbf{x}), \hat{\mathbf{a}}_0\rangle - f^*(\mathbf{x})\big)^2$.
- Risk of *trained* CK (after one step): $\mathcal{R}_{\mathrm{GD}}(\lambda) = \mathbb{E}_{\mathbf{x}}\big(\langle \sigma(\mathbf{W}_1^\top \mathbf{x}), \hat{\mathbf{a}}_1\rangle - f^*(\mathbf{x})\big)^2$.

Theorem (BES+22)

For $\eta = \Theta(1)$ and $\lambda > 0$, as $n/d \to \psi_1$, $N/d \to \psi_2$, we have

$$\mathcal{R}_0(\lambda) - \mathcal{R}_{\mathrm{GD}}(\lambda) \xrightarrow{\mathbb{P}} \delta(\eta, \lambda, \psi_1, \psi_2).$$

- $\delta(\eta, \lambda, \psi_1, \psi_2)$ *is a* **non-negative** *function of $\eta, \lambda, \psi_1, \psi_2 \in (0, +\infty)$;*
- δ *vanishes if and only if (at least) one of μ_1^*, μ_1 and η is zero.*

Provable improvement over the initial CK model!

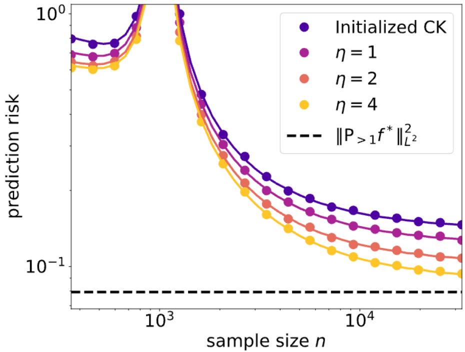
Note: this does not require the student and teacher to have the same nonlinearity.

Small Learning Rate (Continued)

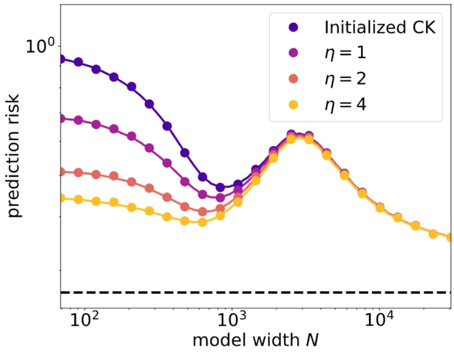
In some special cases, the expression of δ can be further simplified.

Proposition (BES+22)

- **[Large sample limit]** *As* $\psi_1 \to \infty$, δ *is* **increasing** *with respect to* η.
- **[Large width limit]** *As* $\psi_2 \to \infty$, $\delta(\eta, \lambda, \psi_1, \psi_2) \to 0$.



Risk vs. sample size.



Risk vs. model width.

Note: In all cases, $R_0(\lambda) \geq R_{\mathrm{GD}}(\lambda) \geq \|P_{>1} f^*\|_{L^2}^2$ due to the GET under $\eta = \Theta(1)$.

Analysis of Large Learning Rate ($\eta = \Theta(\sqrt{d})$)

Finally, we consider the large learning rate regime with $\eta = \Theta(\sqrt{d})$.

- $\mathbf{W}_1$ travels far away from initialization $\Rightarrow$ CK can be "nonlinear" ☺
- In the absence of GET, precise analysis of prediction risk is difficult ☹

Alternative: upper-bound $\mathcal{R}_{\mathrm{GD}}(\lambda)$ and compare against *kernel lower bound*.

We define: $\tau^* := \inf_\eta \mathbb{E}_{\xi_1}\left[\sigma_*(\xi_1) - \mathbb{E}_{\xi_2}\left(\sigma(\eta\xi_1 + \xi_2)\right)\right]^2$

Lemma (BES+22)

[Informal] Given **bounded** *activation σ, after one GD step on $\mathbf{W}$ with $\eta = \Theta(\sqrt{N})$, there exists some $\tilde{f}(\mathbf{x}) = \frac{1}{\sqrt{N}}\tilde{\mathbf{a}}^\top \sigma(\mathbf{W}_1^\top \mathbf{x})$ that achieves prediction risk "close" to τ^*.*

- τ^* can be interpreted as some measure of "model misspecification".
- **Note:** the definition of τ^* does *not* involve the specific value of step size η.

Large Learning Rate (Continued)

Theorem (BES+22)

After one GD step on W with $\eta = \Theta(\sqrt{N})$, there exist constants $C, \psi_1^* > 0$ such that for any $\psi_1 > \psi_1^*$, and $n^{-1+\epsilon} < N^{-1}\lambda < n^{-\epsilon}$ for some small $\epsilon > 0$,

$$R_{\mathrm{GD}}(\lambda) \;\leq\; 16\tau^* + C\sqrt{\tau^*} \cdot \psi_1^{-1/2} + \psi_1^{-1},$$

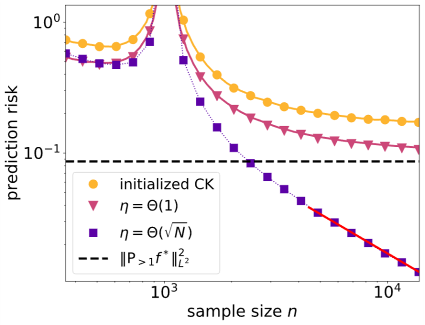
with probability 1, as $n, d, N \to \infty$ proportionally.

If $\tau^* \ll \|P_{>1} f^*\|_{L^2}^2$, CK ridge regression after one feature learning step **outperforms the kernel ridge lower bound**:

- $\sigma = \sigma_* = \tanh$: $\quad R_{\mathrm{GD}}(\lambda) < \|P_{>1} f^*\|_{L^2}^2$
- $\sigma = \sigma_* = \mathrm{erf}$: $\quad$ there exists constant $C > 0$ s.t. $R_{\mathrm{GD}}(\lambda) \leq C \cdot \psi_1^{-1} = \Theta(d/n)$

Caution: separation only present in specific (σ, σ_*).

$\sigma = \sigma_* = \mathrm{erf}$, $\;\eta = N^\alpha$, $\alpha \in [0, 1/2]$.



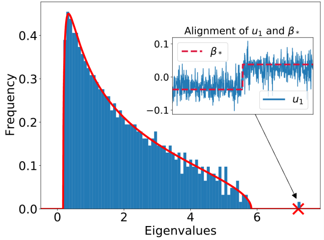
Summary of Results

How Does One Gradient Step Change the Weights?

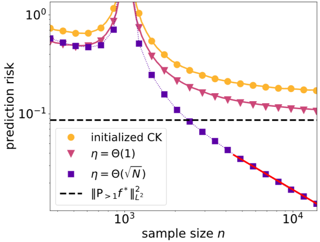
- The isolated singular vector of W^1 aligns with the *linear component* of f^*.
- The top eigenvector of the CK matrix aligns with training labels y (conjecture).

How Do the Learned Features Improve Generalization?

- $\eta = \Theta(1)$ – **Linear Regime.** Precise analysis via GET; $R_0 \geq R_{\text{GD}} \geq \|P_{>1} f^*\|_{L^2}^2$.
- $\eta = \Theta(\sqrt{d})$ – **Nonlinear Regime.** For certain f^*, $R_0 \geq \|P_{>1} f^*\|_{L^2}^2 \geq R_{\text{GD}}$.



Signal+noise structure of W^1.



Improvement of prediction risk.

References

- El Karoui, 2010. The spectrum of kernel random matrices.
- Louart, Liao, and Couillet, 2018. A random matrix approach to neural networks.
- Mei and Montanari, 2019. The generalization error of random features regression: Precise asymptotics and double descent curve.
- Ghorbani et al., 2019. Linearized two-layer neural networks in high dimensions.
- Hu and Lu, 2020. Universality laws for high-dimensional learning with random features.
- Fan and Wang, 2020. Spectra of the conjugate kernel and neural tangent kernel for linear-width neural networks.
- Gerace et al., 2020. Generalisation error in learning with random features and the hidden manifold model.
- Yang and Hu, 2021. Feature learning in infinite-width neural networks.
- Loureiro et al., 2021. Learning curves of generic features maps for realistic datasets with a teacher-student model.
- Bartlett et al., 2021. Deep learning: a statistical viewpoint.

Large Sample Covariance Matrices without Independence Structures in Columns

Bai and Zhou, Statistica Sinica 18 (2008)

Presentation for Random Matrix Theory

Presentation order

Notation $\rightarrow$ literature review $\rightarrow$ motivating examples $\rightarrow$ main theorem $\rightarrow$ example consequences $\rightarrow$ proof sketch.

Notation preliminary: data matrix and sample covariance

For each n, let $p = p(n)$ and let

$$\mathcal{X}_n = (X_{jk})_{1\le j\le p,\,1\le k\le n} = (\mathbf{X}_1, \dots, \mathbf{X}_n) \in \mathbb{C}^{p\times n}.$$

Here

$$X_{jk} = j\text{-th coordinate of the } k\text{-th observation}, \qquad \mathbf{X}_k = (X_{1k}, \dots, X_{pk})^T \in \mathbb{C}^p.$$

Dependence structure

The columns $\mathbf{X}_1, \dots, \mathbf{X}_n$ are independent, but within a fixed column the coordinates

$$X_{1k}, \dots, X_{pk}$$

may be dependent.

Sample covariance matrix

$$B_n = \frac{1}{n}\mathcal{X}_n\mathcal{X}_n^* \in \mathbb{C}^{p\times p}, \qquad \frac{p}{n} \to c \in (0, \infty).$$

Here * means conjugate transpose. In real-valued examples, * is just transpose.

Notation preliminary: spectral distribution and covariance profile

Let $\lambda_1, \dots, \lambda_p$ be the eigenvalues of B_n. The empirical spectral distribution is

$$F^{B_n}(x) = \frac{1}{p} \sum_{j=1}^{p} \mathbf{1}(\lambda_j \leq x).$$

Its Stieltjes transform is

$$m_n(z) = \int \frac{1}{x - z}\, dF^{B_n}(x) = \frac{1}{p} \operatorname{tr}(B_n - zI_p)^{-1}, \qquad z \in \mathbb{C}^+.$$

Column covariance profile

$$T_n = \mathbb{E}(\mathbf{X}_k \mathbf{X}_k^*) = (t_{j\ell})_{1 \leq j, \ell \leq p}, \qquad t_{j\ell} = \mathbb{E}\big(X_{jk} \overline{X_{\ell k}}\big).$$

Goal: describe the limit of F^{B_n} from T_n and weak dependence conditions.

Literature review I: iid entries and latent iid structure

Classical Marčenko–Pastur line

For iid entries and $p/n \to c$, the Empirical spectral distribution (ESD) of $n^{-1}XX^*$ converges to the MP law:

$$F^{\frac{1}{n}XX^*} \Rightarrow \mathrm{MP}_c\,.$$

Extensions include Wachter, Yin, and Bai's review.

Silverstein population-covariance model

A major extension allows a deterministic covariance profile:

$$X_k = T_n^{1/2} Y_k,$$

where Y_k has iid entries. This leads to a generalized MP/Silverstein equation depending on

$$F^{T_n} \Rightarrow H.$$

Gap: Spearman rank vectors and self-normalized sample-correlation columns do not naturally fit $T_n^{1/2} Y_k$ with iid entries.

Literature review II: isotropy and the remaining gap

Isotropic-vector models

Yin–Krishnaiah and Bai–Yin–Krishnaiah considered vectors with strong symmetry, roughly

$$UX_k \overset{d}{=} X_k \quad \text{for every orthogonal/unitary } U.$$

This allows dependent coordinates, but only under rotational symmetry.

Why the applications still do not fit

Rank vectors, normalized columns, finite-population samples, and AR(1) time series are not rotationally symmetric.

Bai–Zhou's contribution

They keep independent columns, but replace entrywise independence or isotropy by a direct quadratic-form concentration condition.

Literature review III: sample-correlation moment assumptions

Earlier benchmark: Jiang (2004)

Spectral distributions of large sample correlation matrices were studied under second-moment assumptions. In the iid scalar setup, this means roughly

$$\mathbb{E}Y^2 < \infty.$$

Bai–Zhou's sample-correlation improvement

In Section 2.2, Bai–Zhou replace finite variance by the weaker condition

$$Y \in \mathcal{D}(N),$$

meaning Y is in the *normal domain of attraction*: for iid copies $Y_1, Y_2, \ldots$, there exist $a_m \in \mathbb{R}$ and $b_m > 0$ such that

$$\frac{Y_1 + \cdots + Y_m - a_m}{b_m} \Rightarrow N(0, 1).$$

$$\mathbb{E}Y^2 < \infty \quad \Longrightarrow \quad Y \in \mathcal{D}(N), \qquad \text{but not conversely.}$$

Thus Bai–Zhou allow some heavy-tailed cases beyond Jiang's finite-variance setting.

Benchmark: the Marčenko–Pastur law

Classical iid-entry covariance matrix

If the entries are iid with mean 0, variance 1, and $p/n \to c$, then

$$F^{\frac{1}{n}\mathcal{X}_n\mathcal{X}_n^*} \Rightarrow \mathrm{MP}_c\,.$$

The MP density is

$$f_c(x) = \frac{1}{2\pi c x}\sqrt{(b-x)(x-a)}\,\mathbf{1}_{[a,b]}(x), \quad a = (1-\sqrt{c})^2, \quad b = (1+\sqrt{c})^2.$$

Stieltjes transform

$$m_{\mathrm{MP},c}(z) = \frac{1-c-z+\sqrt{(1+c-z)^2-4c}}{2cz}.$$

MP is the default bulk spectrum of high-dimensional noise.

Motivating example I: Spearman rank correlation matrix

Suppose $\{Y_{jk} : 1 \le j \le p,\ 1 \le k \le n\}$ are independent with a continuous distribution. Here j indexes the variable/row and k indexes the observation/sample. Let

$$Q_{jk} = \text{rank of } Y_{jk} \text{ among } Y_{j1}, \ldots, Y_{jn}.$$

For each variable j, define the centered rank vector

$$Z_j = \frac{\sqrt{12p}}{\sqrt{n(n^2-1)}} \left(Q_{j1} - \frac{n+1}{2}, \ldots, Q_{jn} - \frac{n+1}{2} \right)^T \in \mathbb{R}^n.$$

Set

$$Z_p = (Z_1, \ldots, Z_p) \in \mathbb{R}^{n \times p}.$$

Then the Spearman rank correlation matrix is the Gram matrix

$$R_p = \frac{1}{p} Z_p^T Z_p \in \mathbb{R}^{p \times p}.$$

Spearman: where dependence enters

For each fixed variable j, the ranks

$$Q_{j1}, \ldots, Q_{jn}$$

are a permutation of $1, \ldots, n$. Therefore the centered rank coordinates satisfy deterministic constraints:

$$\sum_{k=1}^{n} \left(Q_{jk} - \frac{n+1}{2} \right) = 0,$$

$$\sum_{k=1}^{n} \left(Q_{jk} - \frac{n+1}{2} \right)^2 = \frac{n(n^2 - 1)}{12}.$$

So the coordinates inside a rank vector cannot be independent.

Classical iid-entry covariance theory no longer applies directly.

Motivating example II: sample correlation matrix

Let

$$Y_n = (Y_{jk})_{1\le j\le p,\,1\le k\le n} = (Y_1,\dots,Y_n) \in \mathbb{R}^{p\times n},$$

where the columns Y_k are iid and $p/n \to c$. In Section 2.2, the paper views Y_k as the p observations of the k-th component/variable.

Normalize each variable before forming correlations

Define

$$\mathbf{X}_k = \sqrt{n}\,\frac{Y_k}{\|Y_k\|}, \qquad \mathcal{X}_n = (\mathbf{X}_1,\dots,\mathbf{X}_n).$$

Then the sample correlation matrix is

$$R_Y = \frac{1}{n}\mathcal{X}_n^T\mathcal{X}_n \in \mathbb{R}^{n\times n}.$$

Indeed, R_Y has diagonal entries 1 because $\|\mathbf{X}_k\|^2 = n$.

Sample correlation: where dependence enters

Even if the raw entries $Y_{1k}, \ldots, Y_{pk}$ are iid, the normalized coordinates

$$X_{jk} = \sqrt{n}\,\frac{Y_{jk}}{\|Y_k\|}$$

inside one column are not independent, because they share the same random denominator. They also satisfy the deterministic constraint

$$\sum_{j=1}^{p} X_{jk}^2 = n.$$

Classical covariance theory for iid entries does not directly apply after self-normalization.

Statistical point: correlation removes marginal scale, so the spectrum of R_Y gives a scale-free high-dimensional noise benchmark.

Main theorem: assumptions and interpretation

Assumptions

As $n \to \infty$,

$$\frac{p}{n} \to c \in (0, \infty), \qquad \|T_n\| = O(1), \qquad F^{T_n} \Rightarrow H.$$

For every deterministic bounded $p \times p$ matrix A,

$$\mathbb{E}\left|\mathbf{X}_k^* A \mathbf{X}_k - \operatorname{tr}(AT_n)\right|^2 = o(n^2).$$

Interpretation

The random quadratic form

$$\mathbf{X}_k^* A \mathbf{X}_k$$

behaves like its covariance trace

$$\operatorname{tr}(AT_n).$$

This is the replacement for entrywise independence inside columns.

Main result: limiting equation

Fixed-point equation

The limiting Stieltjes transform $m(z)$ satisfies

$$m(z) = \int \frac{1}{t\{1 - c - czm(z)\} - z} \, \mathrm{d}H(t), \qquad z \in \mathbb{C}^+.$$

Equivalently, for the companion transform

$$\underline{m}(z) = -\frac{1-c}{z} + cm(z),$$

we have

$$z = -\frac{1}{\underline{m}(z)} + c \int \frac{t}{1 + t\underline{m}(z)} \, \mathrm{d}H(t).$$

MP versus non-MP

If $H = \delta_1$, the result is MP. If H is non-degenerate, the result is generally a deformed, non-MP limit.

Corollary 1.1 as an application tool

The theorem's quadratic-form condition can be hard to check directly:

$$\mathbb{E}|\mathbf{X}_k^* A \mathbf{X}_k - \mathrm{tr}(AT_n)|^2 = o(n^2).$$

Corollary 1.1 replaces it by low-order mixed moment checks such as controlling

$$\mathbb{E}\left(\overline{X}_{jk} X_{lk} - t_{lj}\right)\left(X_{j'k}\overline{X}_{l'k} - t_{j'l'}\right).$$

Spearman after the theorem: conclusion

Corollary 1.1 reduces the abstract condition to low-order moment estimates. Bai–Zhou compute, for example,

$$\mathbb{E}Z_{11}^4 \sim \frac{18p^2}{5n^2}, \qquad \mathbb{E}Z_{11}^2 = \frac{p}{n},$$
$$\mathbb{E}Z_{11}^2 Z_{21} Z_{31} \sim -\frac{p^2}{n^3}, \qquad \mathbb{E}Z_{11}Z_{21}Z_{31}Z_{41} \sim \frac{3p^2}{n^4},$$
$$\mathbb{E}\Big[(Z_{11}^2 - \mathbb{E}Z_{11}^2)(Z_{21}^2 - \mathbb{E}Z_{21}^2)\Big] = O(n^{-1}).$$

These estimates verify Corollary 1.1.

Theorem 2.2: Spearman rank correlation matrix

If $p/n \to c$, then

$$F^{R_p} \Rightarrow \mathrm{MP}_c \quad \text{a.s.}, \qquad m_{R_p}(z) = \frac{1 - c - z + \sqrt{(1 + c - z)^2 - 4c}}{2cz}.$$

Sample correlation after the theorem: fit and covariance profile

Matrix form. Define

$$\mathcal{X}_n = (\mathbf{X}_1, \ldots, \mathbf{X}_n), \qquad \mathbf{X}_k = \sqrt{n}\,\frac{Y_k}{\|Y_k\|} \in \mathbb{R}^p.$$

Apply the theorem to

$$B_n = \frac{1}{n}\mathcal{X}_n\mathcal{X}_n^T \in \mathbb{R}^{p\times p}, \qquad p/n \to c.$$

The target correlation matrix is the Gram matrix

$$R_Y = \frac{1}{n}\mathcal{X}_n^T\mathcal{X}_n \in \mathbb{R}^{n\times n},$$

which has the same nonzero eigenvalues as B_n.

Covariance profile. Self-normalization creates

$$\sum_{j=1}^{p} X_{jk}^2 = n, \qquad \mathbb{E}X_{1k}^2 = \frac{n}{p} \to \frac{1}{c}.$$

Bai–Zhou show

$$T_n = \mathbb{E}(\mathbf{X}_k\mathbf{X}_k^T), \qquad \|T_n\| = O(1), \qquad F^{T_n} \Rightarrow \delta_{1/c}.$$

Sample correlation after the theorem: conclusion

Corollary 1.1 is verified using moment estimates for self-normalized sums. The paper quotes estimates from Giné–Götze–Mason:

$$n^{-2}\mathbb{E}X_{11}^4 = o(p^{-1}), \qquad n^{-2}\mathbb{E}X_{11}^2 X_{21} X_{31} = o(p^{-3}),$$
$$n^{-2}\mathbb{E}X_{11}X_{21}X_{31}X_{41} = o(p^{-4}), \qquad n^{-2}\mathbb{E}X_{11}^2 X_{21}^2 = O(p^{-2}).$$

These estimates control the quadratic-form fluctuations required by the theorem.

Theorem 2.3: sample correlation matrix

If Y belongs to the domain of attraction of the normal law and $p/n \to c$, then

$$F^{R_Y} \Rightarrow \mathrm{MP}_{1/c} \quad \text{a.s.},$$

with Stieltjes transform

$$m_{R_Y}(z) = \frac{-(cz - c + 1) + \sqrt{(cz - c - 1)^2 - 4c}}{2z}.$$

Sample correlation: what improves over Jiang?

Jiang (2004)

- Object: sample correlation matrices.
- Moment assumption: finite second moment,

$$\mathbb{E}Y^2 < \infty.$$

- Gives an MP-type limiting spectrum for correlations.

Bai–Zhou (2008)

- Same object, but as an application of the dependent-column theorem.
- Assumption: normal domain of attraction,

$$Y \in \mathcal{D}(N).$$

- This can hold even when $\mathbb{E}Y^2 = \infty$.

Proof sketch: same map, weaker input

1. Random part vanishes

$$m_n(z) - \mathbb{E}m_n(z) \to 0 \quad a.s.$$

Use column independence to write the difference as a martingale sum.

2. Mean satisfies equation The resolvent method produces quadratic forms of the type

$$\frac{1}{n}\mathbf{X}_k^* A_k \mathbf{X}_k,$$

where A_k is built from the other columns and is independent of $\mathbf{X}_k$. The assumption replaces this by

$$\frac{1}{n}\operatorname{tr}(A_k T_n).$$

3. Uniqueness closes proof Every subsequential limit solves the same fixed-point equation; uniqueness in $\mathbb{C}^+$ forces one limit.

Conceptual improvement

The proof follows the Bai–Silverstein resolvent architecture, but the iid-entry assumption is replaced by exactly the concentration property the proof needs.

Takeaways

1. The paper studies covariance spectra with dependent coordinates inside columns.
2. Spearman rank matrices and sample correlation matrices motivate the problem.
3. Older iid, latent-iid, and isotropic frameworks do not cover these examples cleanly.
4. Bai–Zhou isolate the key condition: quadratic-form concentration around the covariance trace.

References

Z. Bai and W. Zhou.
Large sample covariance matrices without independence structures in columns.
Statistica Sinica, 18:425–442, 2008.

V. A. Marčenko and L. A. Pastur.
Distribution of eigenvalues for some sets of random matrices.
Mat. Sb., 72:507–536, 1967.

J. W. Silverstein.
Strong convergence of the empirical distribution of eigenvalues of large dimensional random matrices.
J. Multivariate Anal., 55:331–339, 1995.

Learning under Low-Dimensional Structure

A Spiked Random Matrix Perspective

Peng Wang

Based on Jimmy Ba, Murat A. Erdogdu, et al (2023)

April 16, 2026

Industrial and Systems Engineering, USC

Outline

In this talk we will travel from the motivation for studying structured covariances, through two contrasting learning approaches, to the main theoretical results and the future research directions they inspire.

- **Introduction**: problem statement and learning objective
- **Student models**: Kernel ridge regression and two-layer neural networks
- **Main results**: conditions for learnability under a spiked covariance
- **Follow-up ideas**: synthetic data for continual learning under infinite domains
- **Discussion**: limitations, open questions and conclusion

Introduction

Problem statement: a single-index function under a spiked covariance

We consider a high-dimensional random vector $\boldsymbol{x} \in \mathbb{R}^d$ drawn from an anisotropic Gaussian distribution whose covariance matrix contains a single prominent direction. Concretely,

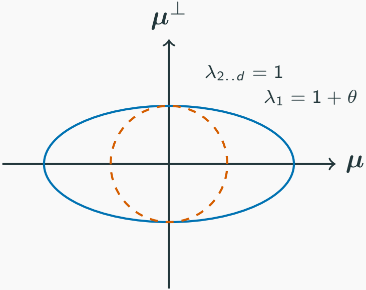
$$y = f_*(\boldsymbol{x}) = \sigma_*\Big(\frac{1}{\sqrt{1+\theta}}\,\langle \boldsymbol{x}, \boldsymbol{\mu}\rangle\Big), \qquad \boldsymbol{x} \sim \mathcal{N}\big(\mathbf{0},\, \boldsymbol{I}_d + \theta\,\boldsymbol{\mu}\,\boldsymbol{\mu}^\top\big),$$

where $\boldsymbol{\mu} \in \mathbb{R}^d$ is a unit vector defining the *spike* and $\theta \gg 0$ controls its strength. The link function $\sigma_* : \mathbb{R} \to \mathbb{R}$ is a fixed polynomial. When $\theta = 0$ the data are isotropic; for $\theta > 0$ a unique direction becomes informative.

Key question: How large must the spike magnitude θ be for simple learners to approximate f_* in high dimension? Our focus is on the proportional limit $n, d \to \infty$ with $n/d \to \psi \in (0, \infty)$.

Interpreting the spiked covariance

The covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{I}_d + \theta\,\boldsymbol{\mu}\,\boldsymbol{\mu}^\top$ has eigenvalues $1+\theta$ along $\boldsymbol{\mu}$ and 1 in orthogonal directions. Thus only the component $\langle \boldsymbol{x}, \boldsymbol{\mu} \rangle$ carries signal. We visualise this by sketching the ellipsoid defined by $\boldsymbol{\Sigma}$.



The rank-one perturbation selects the spike direction and stretches the covariance ellipsoid along it. As θ increases this direction becomes easier to detect and exploit.

Learning objective and proportional regime

We observe n samples $(\boldsymbol{x}_i, y_i)$ and estimate a function $\hat{f}$ to approximate f_*. We measure performance by the prediction risk

$$\mathcal{R}(\hat{f}) = \mathbb{E}_{\boldsymbol{x}}\big[(\hat{f}(\boldsymbol{x}) - f_*(\boldsymbol{x}))^2\big],$$

where the expectation is with respect to the spiked distribution.

We say that a method *learns* f_* in the proportional limit if $\mathcal{R}(\hat{f}) \to 0$ with high probability using only $n \asymp d$ samples.
Our aim is to compare two simple learners under this regime:

1. **Kernel ridge regression** with an inner-product kernel $k(\boldsymbol{x}, \boldsymbol{y}) = g\big(\langle \boldsymbol{x}, \boldsymbol{y}\rangle / d\big)$.
2. **Two-layer neural networks** trained by a two-stage gradient procedure.

We seek to quantify the threshold on θ beyond which these methods achieve vanishing risk.

Student models

Student model I: Kernel ridge regression

The kernel ridge regressor $\hat{f}_{\text{ker}}$ is defined by

$$\hat{f}_{\text{ker}}(\boldsymbol{x}) = k(\boldsymbol{x}, \boldsymbol{X})^\top (\boldsymbol{K} + \lambda \boldsymbol{I})^{-1} \boldsymbol{y},$$

where $\boldsymbol{K}$ is the Gram matrix with entries $K_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$ and $\lambda > 0$ is a ridge parameter controlling complexity.

We work with inner-product kernels of the form $k(\boldsymbol{x}, \boldsymbol{y}) = g\big(\langle \boldsymbol{x}, \boldsymbol{y} \rangle / d\big)$ whose Taylor coefficients are non-negative so that k is positive semi-definite in any dimension.

Intuition: Kernel methods do not learn features; they rely on fixed similarity functions. In high dimension their ability to approximate nonlinear f_* depends critically on the covariance structure and on the degree of the target polynomial.

Student model II: Two-layer neural network

The neural network predictor of width N takes the form

$$f_{\mathrm{NN}}(\boldsymbol{x}) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} a_i \, \sigma\big(\langle \boldsymbol{x}, \boldsymbol{w}_i \rangle + b_i\big),$$

with trainable weights $\boldsymbol{W} = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_N]$ and $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^N$.

The parameters are initialised at random and updated by a two-stage procedure: one gradient step on the first-layer weights $\boldsymbol{W}$, followed by solving a convex problem for the second-layer coefficients $\boldsymbol{a}$. The width grows with dimension as $N = \Omega(d^{\varepsilon})$.

Feature learning: Unlike kernels, neural networks can adapt representations to align with the spike. A single gradient step suffices to align $\boldsymbol{W}$ with $\boldsymbol{\mu}$, thereby overcoming the curse of dimensionality.

Main results

Main result for kernel ridge regression

The analysis projects f_* onto polynomial subspaces via orthogonal projectors $P_{\leq \ell}$ and $P_{>\ell}$. Under mild assumptions on the kernel g and for a fixed degree ℓ, if the spike magnitude scales as $\theta \asymp d^{\beta}$ with

$$\beta \in \left(1 - \frac{1}{\ell}, 1 - \frac{1}{\ell + 1}\right),$$

then, in the proportional limit $n, d \to \infty$, the prediction risk satisfies

$$\mathcal{R}(\hat{f}_{\text{ker}}) - \|P_{>\ell} f_*\|_{L^2}^2 = o_d(1).$$

In particular, when f_* is a degree-p polynomial ($P_{>p} f_* = 0$), kernel ridge regression learns f_* provided

$$\beta > 1 - \frac{1}{p}.$$

Thus the minimum spike strength grows with the inverse of the degree. In **isotropic data** ($\theta = 0$) kernel methods require $n \gg d^p$ samples to learn a degree-p target, which is impossible in the proportional regime when $p > 1$.

Main result for two-layer neural networks

For neural networks with ReLU activation trained by the two-stage procedure, the critical exponent involves the *information exponent* k of σ_*. Let k be the smallest Hermite degree with a nonzero coefficient in the polynomial expansion of σ_*. Under suitable scaling of the width N and learning rate, the network achieves

$$\mathcal{R}(\hat{f}_{\mathrm{NN}}) = o_d(1)$$

with high probability in the proportional limit whenever

$$\beta > 1 - \frac{1}{k}.$$

Because $k \leq p$ (the lowest nonzero Hermite degree is at most the highest polynomial degree), the neural network requires a smaller spike magnitude than kernel ridge regression. This result formalises the intuition that feature learning can leverage low-dimensional structure more effectively than fixed kernels.

Comparing polynomial degree and information exponent

It is instructive to distinguish between the degree p of the polynomial link function and its information exponent k.

- p measures the highest order of nonlinearity present: $\sigma_*(z) = \sum_{m=0}^{p} c_m z^m$ with $c_p \neq 0$.
- k is the smallest Hermite degree $m \geq 1$ such that the projection of σ_* onto the Hermite polynomial h_m is nonzero. Intuitively, k quantifies how soon the signal appears in the Hermite expansion.

Because the Hermite basis orders monomials by degree but excludes the constant term, one always has $k \leq p$. Consequently, neural networks can learn targets with smaller spike magnitudes than kernels: the threshold for the neural network depends on k, while that for the kernel depends on p.

Follow-up research

Synthetic dataset design for continual domain adaptation

The spiked covariance model suggests a natural way to generate synthetic domains for continual pre-training. Instead of one domain, consider an infinite sequence of domains $\{\mathcal{D}_t\}_{t\geq 1}$ with random vectors

$$\boldsymbol{x} \sim \mathcal{N}\big(\mathbf{0},\, \boldsymbol{I}_d + \theta_t\, \boldsymbol{v}_t\, \boldsymbol{v}_t^\top\big), \qquad y = \sigma^*\Big(\frac{\langle \boldsymbol{x}, \boldsymbol{v}_t\rangle}{\sqrt{1+\theta_t\|\boldsymbol{v}_t\|^2}}\Big),$$

where each spike direction $\boldsymbol{v}_t$ lies in the span of latent directions $\boldsymbol{U} = [\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_r]$. By choosing coefficients $\boldsymbol{a}_t$ and spike strengths θ_t one can design a continuum of related domains. Keeping σ^* fixed isolates the effect of shifting input geometry, allowing us to study domain shift independently of task shift. This synthetic framework provides a feasible way for evaluating continual learning methods.

Discussion and conclusion

Limitations and open questions

Although the spiked model illuminates the role of low-dimensional structure, several questions remain open:

Alignment: The theory assumes that the spike direction exactly matches the teacher's signal. In reality, signals may be misaligned or spread across multiple directions. How does learnability degrade as the alignment decreases, and is there a phase transition?

Training dynamics: The neural network theorem relies on a one-step gradient update of the first layer. Does the threshold $1 - 1/k$ remain sharp under full gradient descent or different optimisers?

Limitations and open questions

Although the spiked model illuminates the role of low-dimensional structure, several questions remain open:

Activation functions: The proof requires random biases to diversify the ReLU activation. Which activations, bias distributions or architectural choices are necessary to realise the separation between kernels and neural networks?

Multi-index teachers: Extending the theory to multiple latent directions or higher-rank spikes could model more complex problems. How does the advantage of neural networks scale with the number of informative components?

Conclusion

The spiked random matrix perspective reveals when simple learning methods can overcome the curse of dimensionality in high dimensions.
Kernel ridge regression, relying on fixed features, requires the spike magnitude to grow like $d^{1-1/p}$ to learn a degree-p polynomial.
Two-layer neural networks, by adapting their representation, succeed under a weaker condition $\beta > 1 - 1/k$, highlighting the power of feature learning.
These results not only sharpen our theoretical understanding but also inspire synthetic benchmarks and continual learning strategies based on low-dimensional structure.

Thank you for your attention! Any questions?